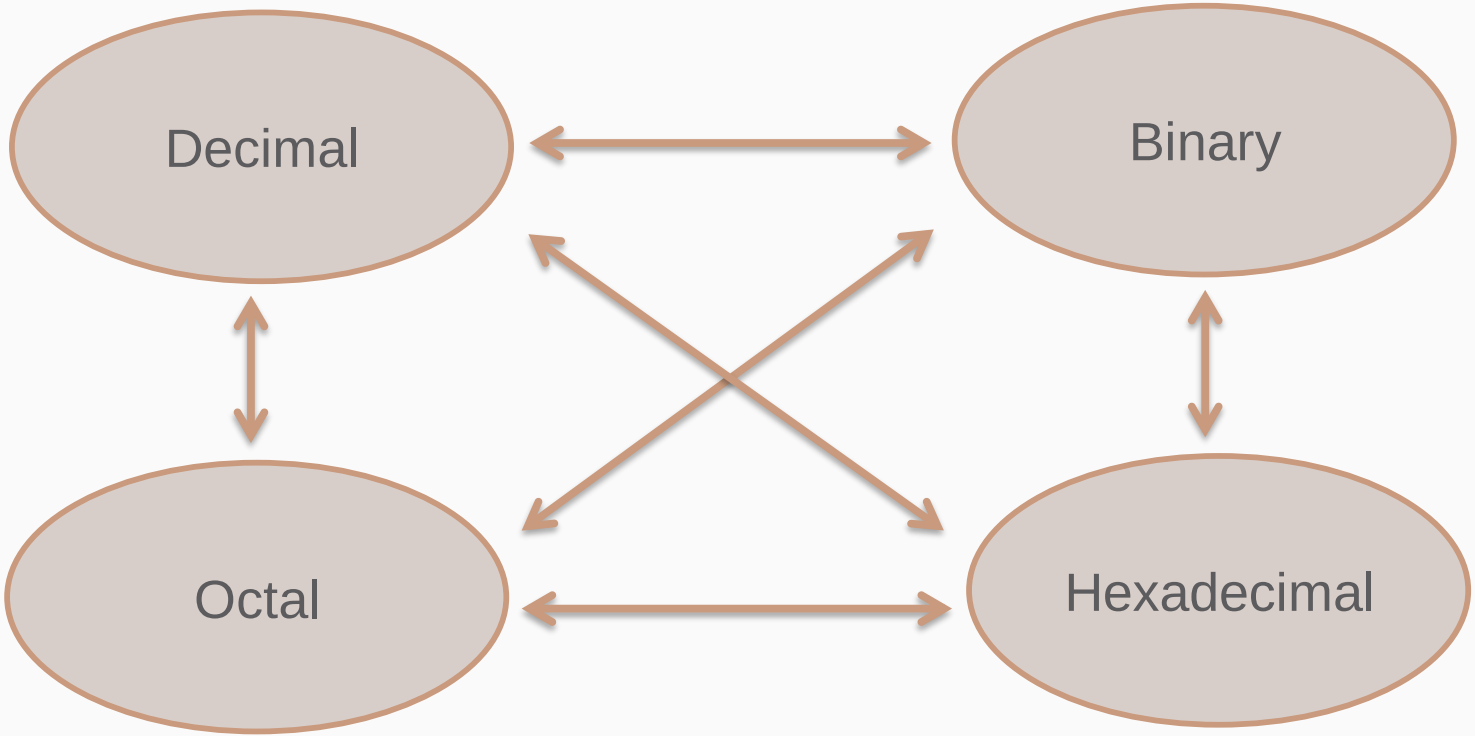
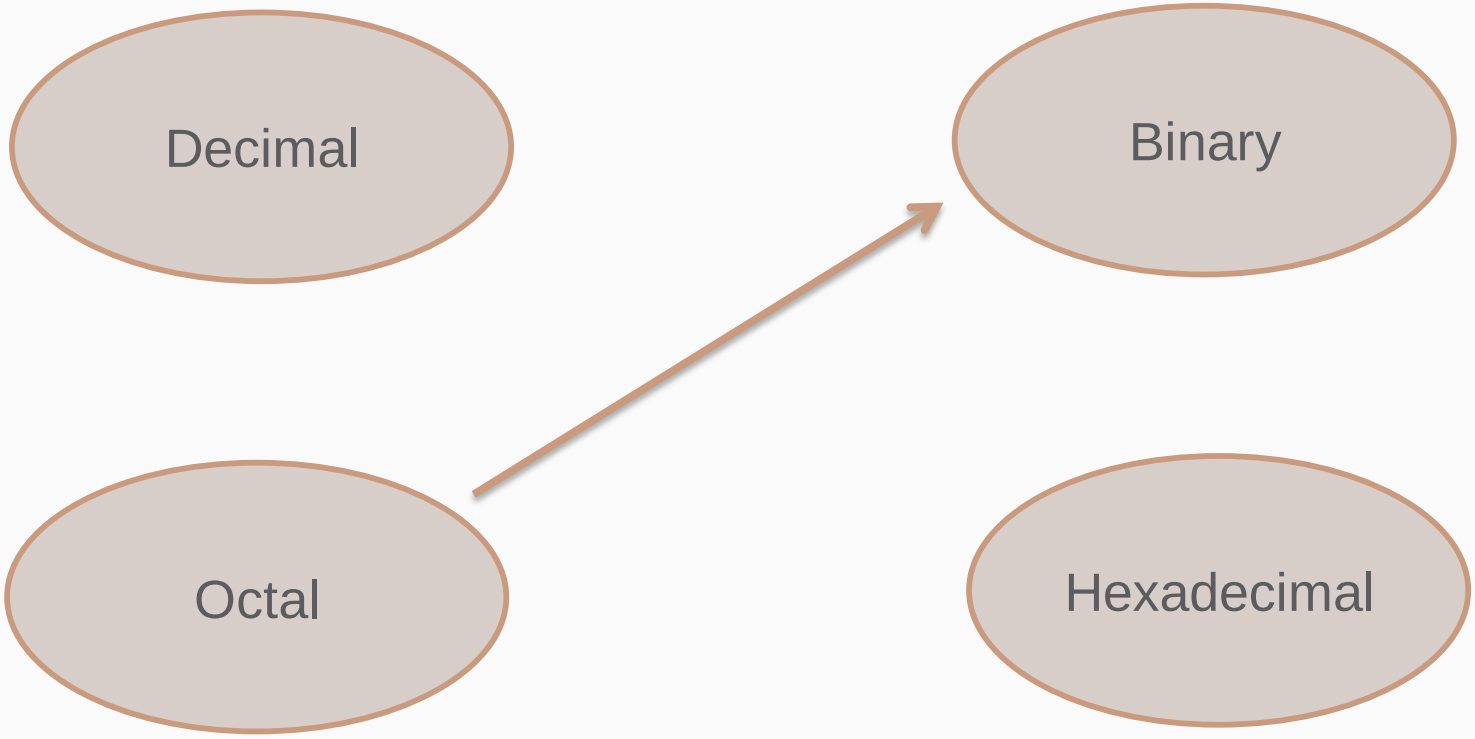
Number System

Conversion Among Bases



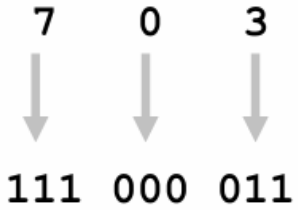
Octal to Binary



Technique

- Convert each octal digit to a 3-bit equivalent binary representation

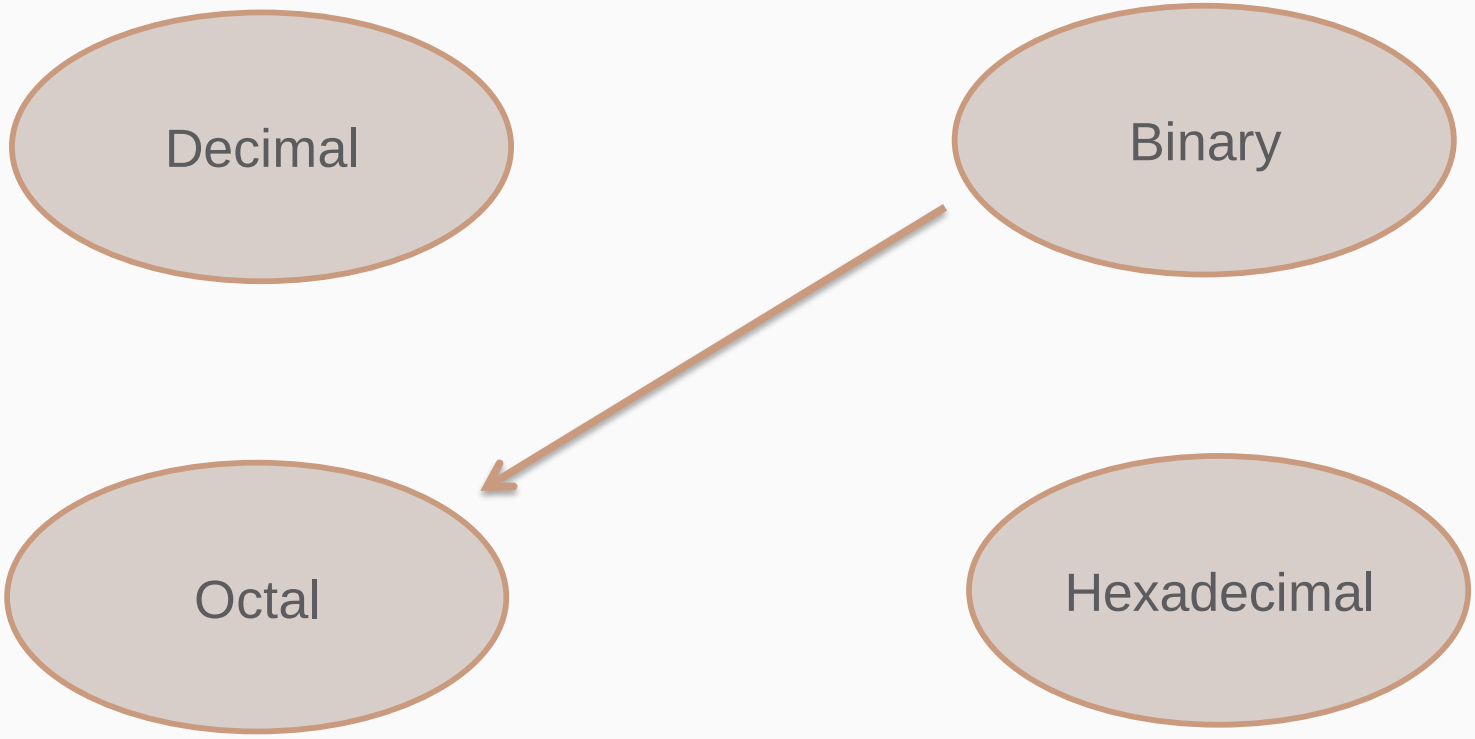
$703_8 = ?_2$



$703_8 = 111000011_2$

- Try to solve :
  - $(213)_8$
  - $(536)_8$
  - $(4321)_8$

Binary to Octal



Technique

- Group bits in threes, starting on right
- Convert to octal digits

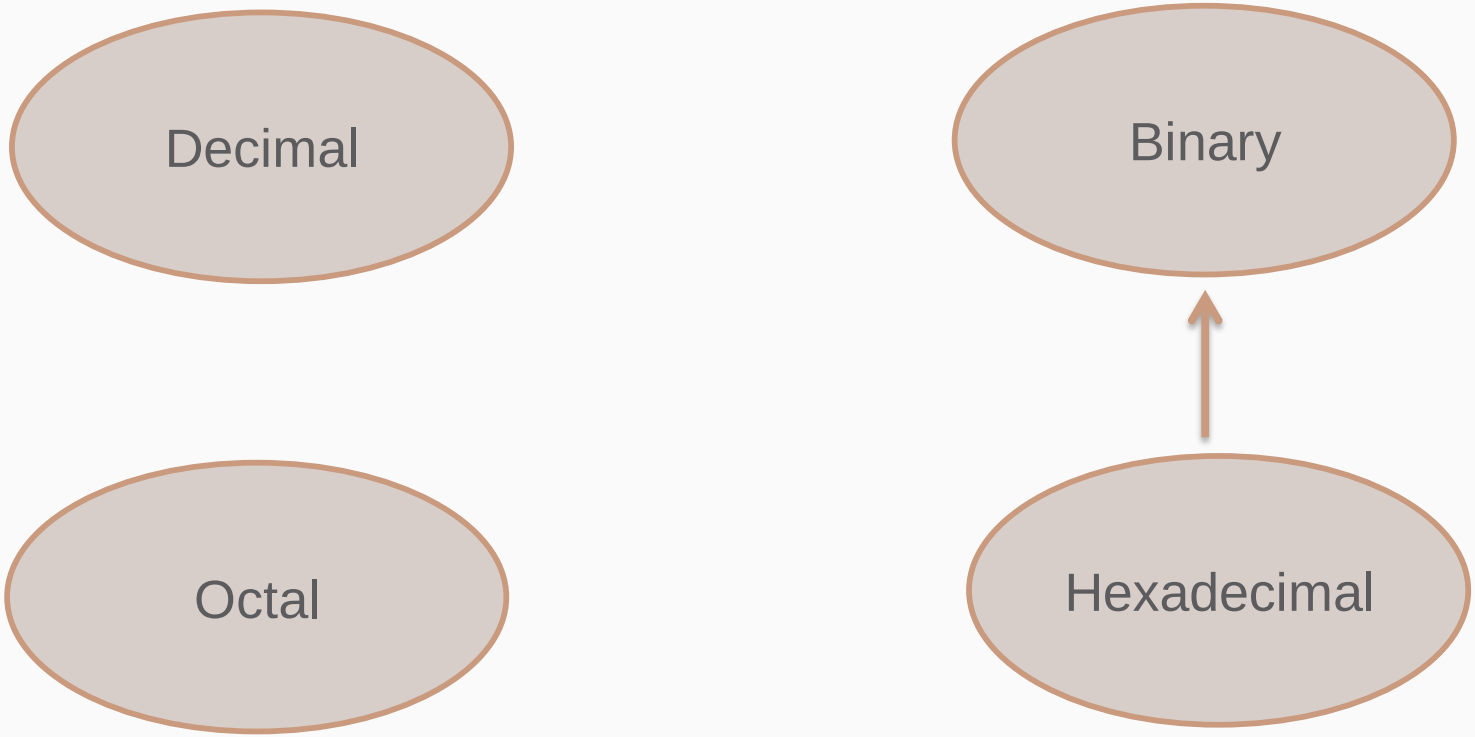
$1011010111_2 = ?_8$



- Try to solve :
  - $(101110001101)_2$
  - $(110011010100)_2$
  - $(11100110001)_2$

$1011010111_2 = 1327_8$

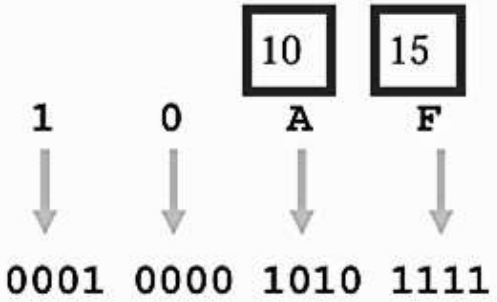
Hexadecimal to Binary



Technique

- Convert each hexadecimal digit to a 4-bit equivalent binary representation

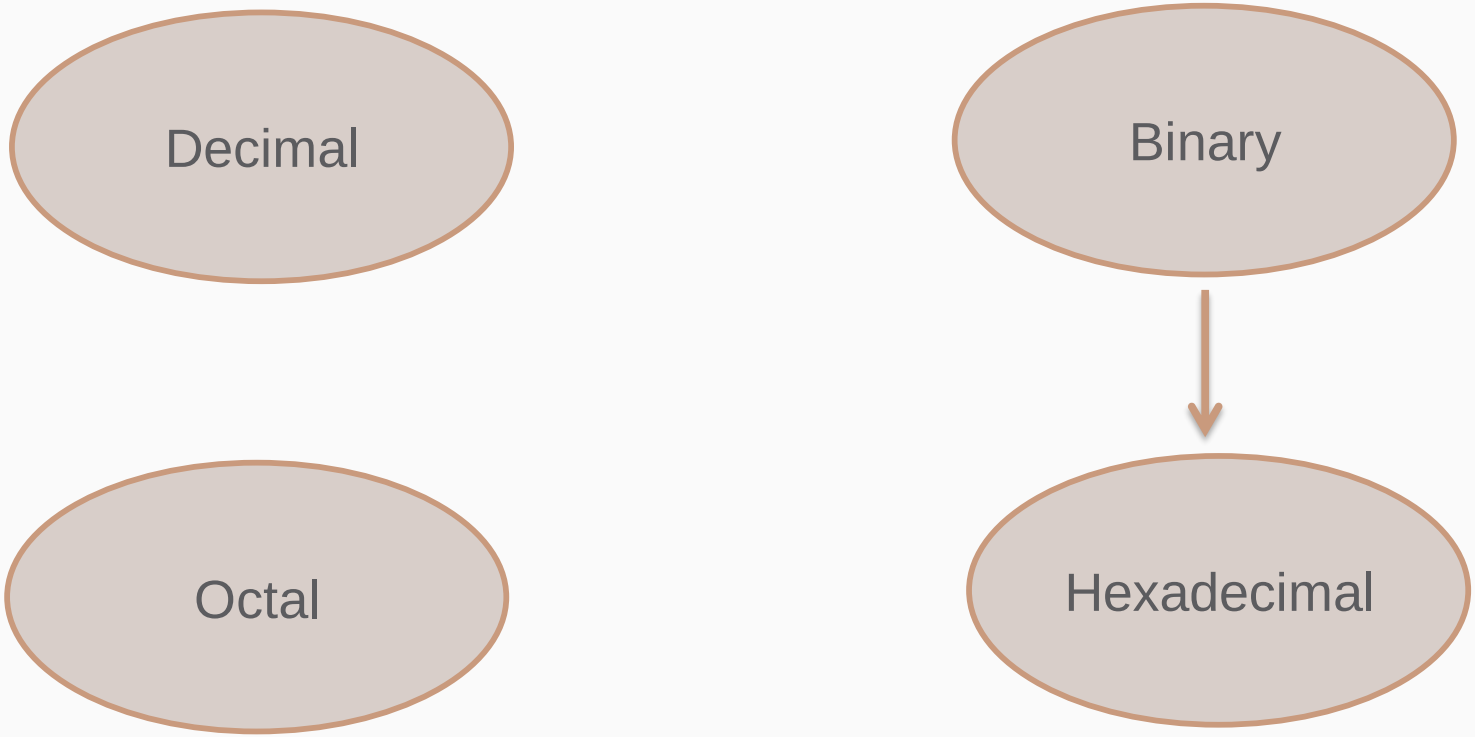
$10AF_{16} = ?_2$



$10AF_{16} = 0001000010101111_2$

- Try to solve :
  - $(8DB4)_{16}$
  - $(9D01)_{16}$
  - $(FD80)_{16}$

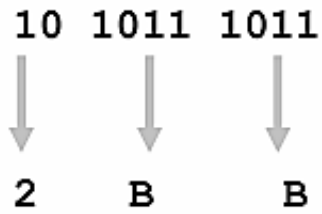
Binary to Hexadecimal



Technique

- Group bits in fours, starting on right
- Convert to hexadecimal digits

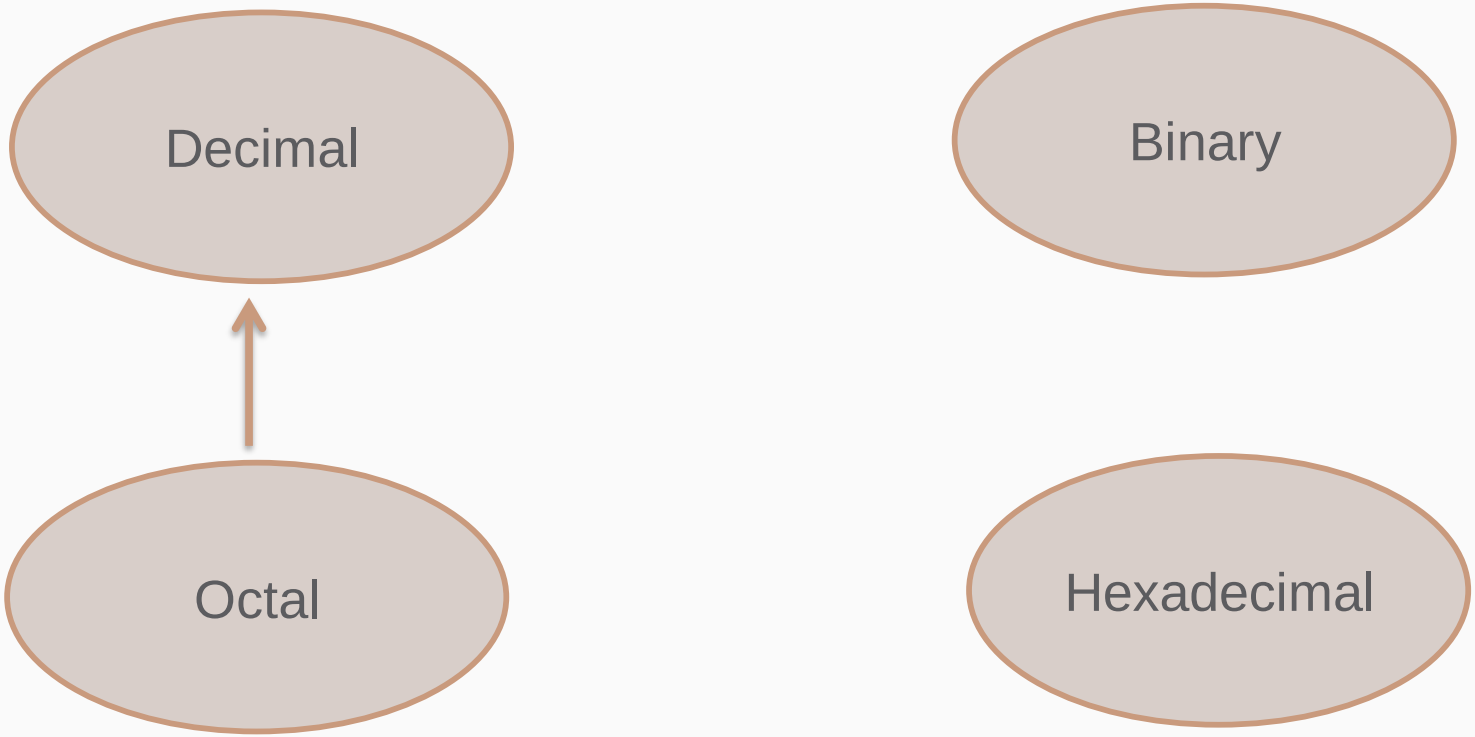
$1010111011_2 = ?_{16}$



$1010111011_2 = 2BB_{16}$

- Try to solve :
  - $(1010101101001)_2$
  - $(1111010100111110)_2$
  - $(110100100111)_2$

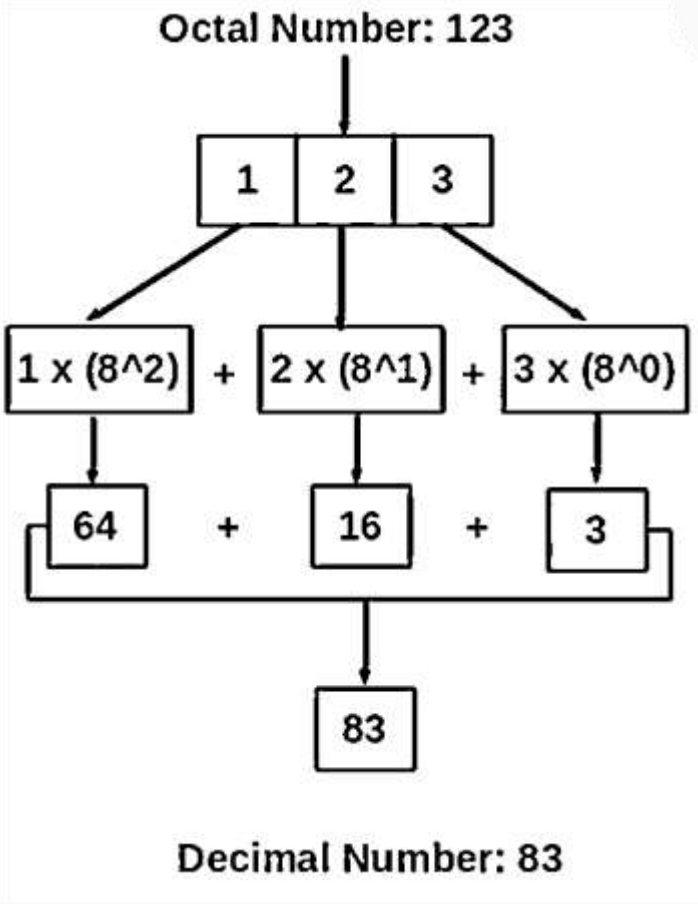
Octal to Decimal



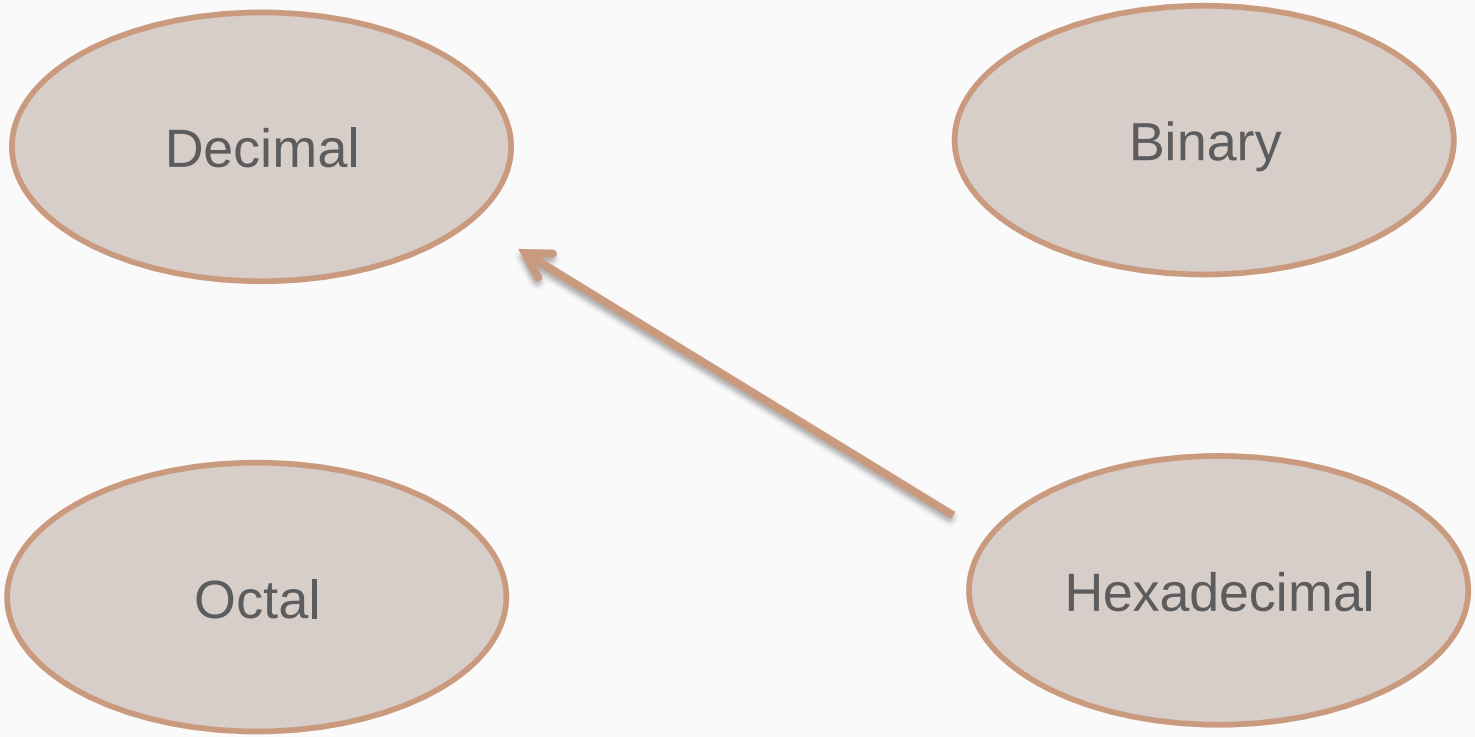
Technique

- Multiply each bit by 8n, where n is the “weight” of the bit
- The weight is the position of the bit, starting from 0 on the right
- Add the results

- Try to solve :
  - $(67)_8$
  - $(341)_8$
  - $(5312)_8$



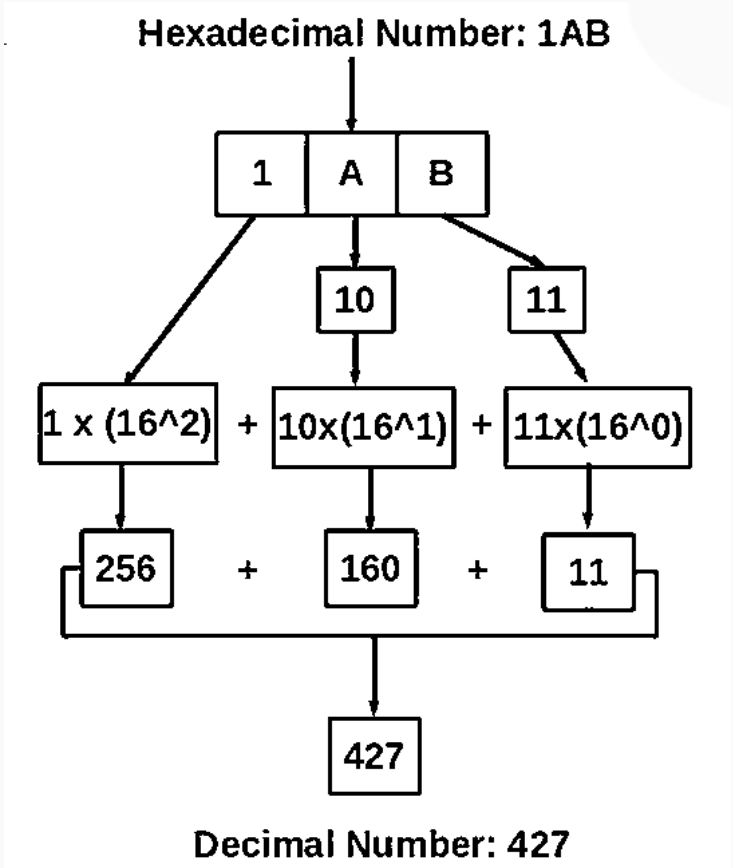
Hexadecimal to Decimal



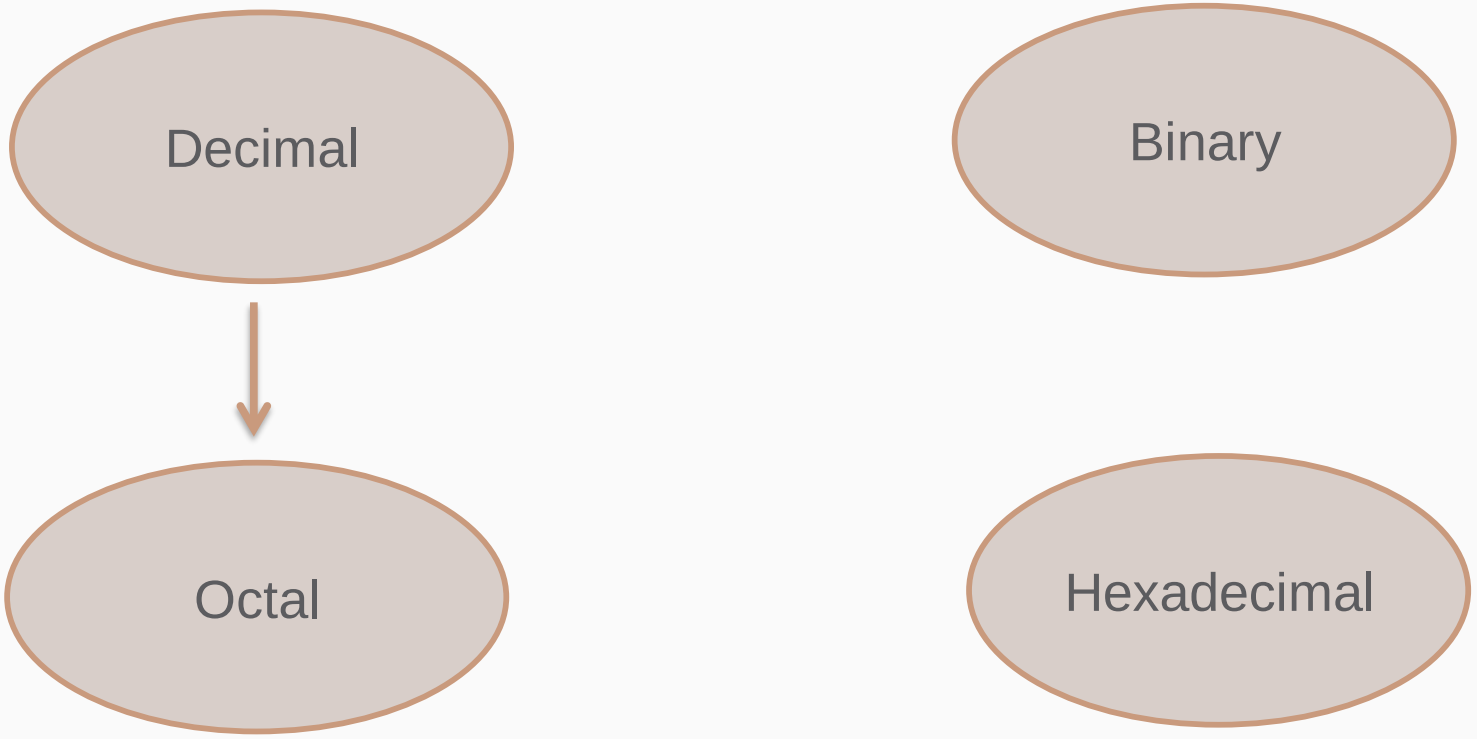
Technique

- Multiply each bit by 16n, where n is the “weight” of the bit
- The weight is the position of the bit, starting from 0 on the right
- Add the results

- Try to solve :
  - $(1A)_{16}$
  - $(512)_{16}$
  - $(F1C)_{16}$



Decimal to Octal



Technique

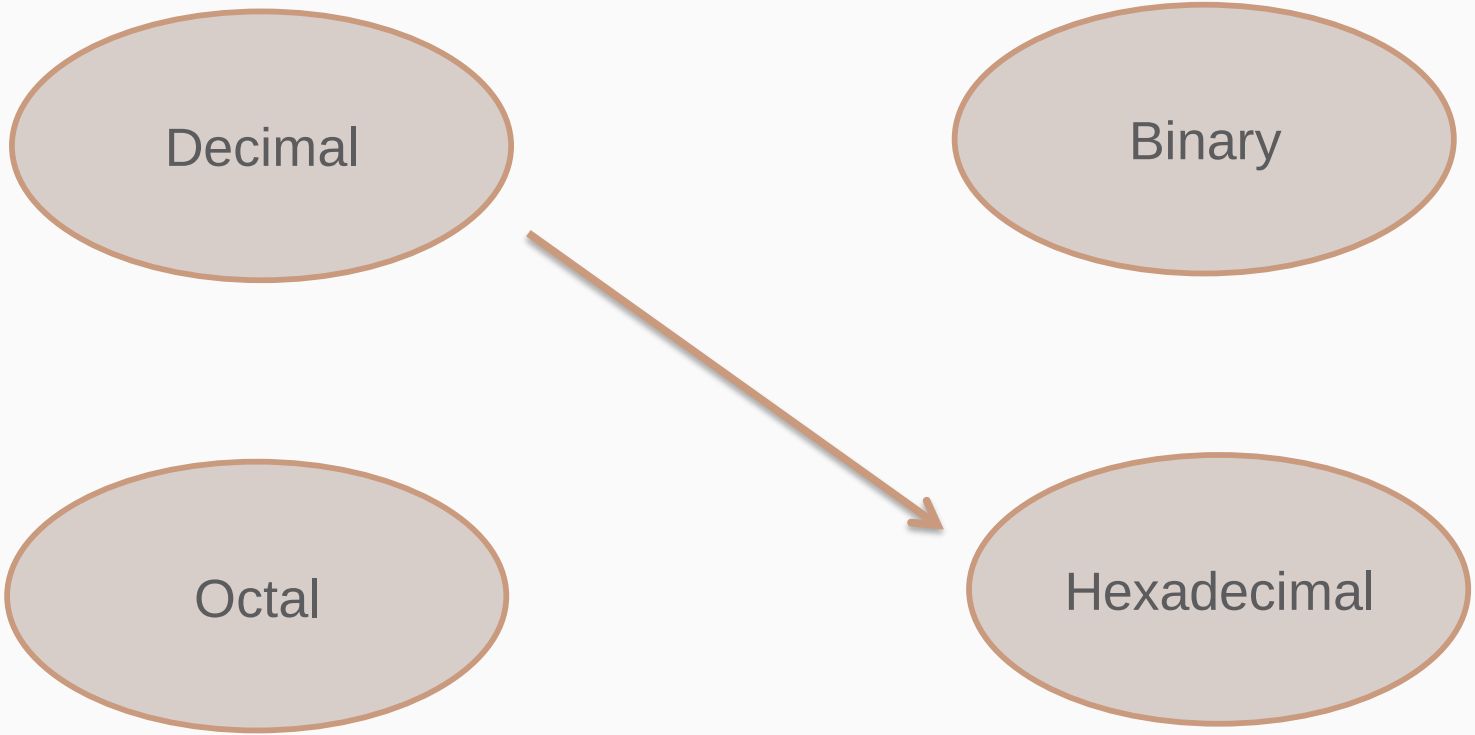
- Divide by 8
- Keep track of the remainder

- Given: Decimal Number = $(775)_{10} = (1407)_8$

- Try to solve :
  - $(456)_{10}$
  - $(500)_{10}$
  - $(1792)_{10}$

Divide by 8	Result	Octal Value (Reminder)
755 / 8	96	7 (LSB)
96 / 8	12	0
12 / 8	1	4
1 / 8	0	1 (MSB)

Decimal to Hexadecimal



Technique

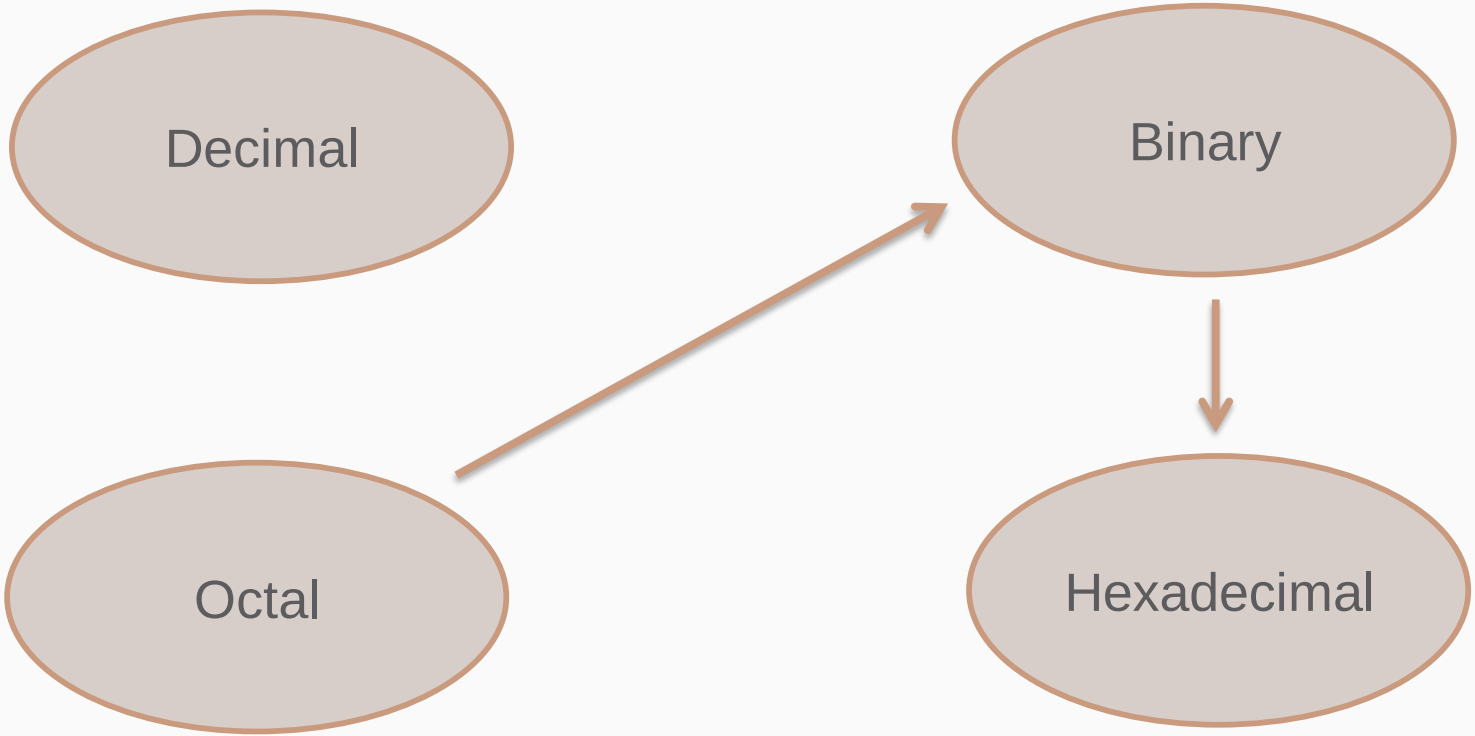
- Divide by 16
- Keep track of the remainder

- Given: Decimal Number = $(284)_{10} = (11C)_{16}$

Divide by 16	Result	Hexadecimal Value (Reminder)
284 / 16	17	12 = C (LSB)
17 / 16	1	1
1 / 16	0	1 (MSB)

- Try to solve :
  - $(34)_{10}$
  - $(160)_{10}$
  - $(1228)_{10}$

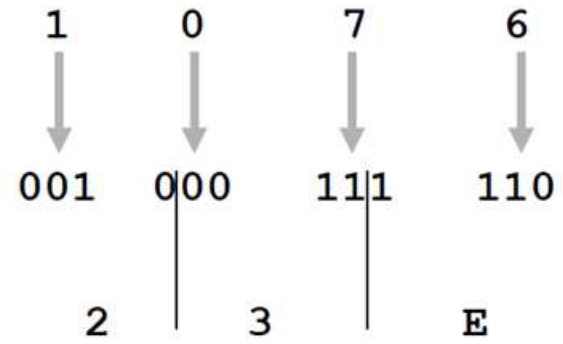
Octal to Hexadecimal



Technique

- Use binary as an intermediary

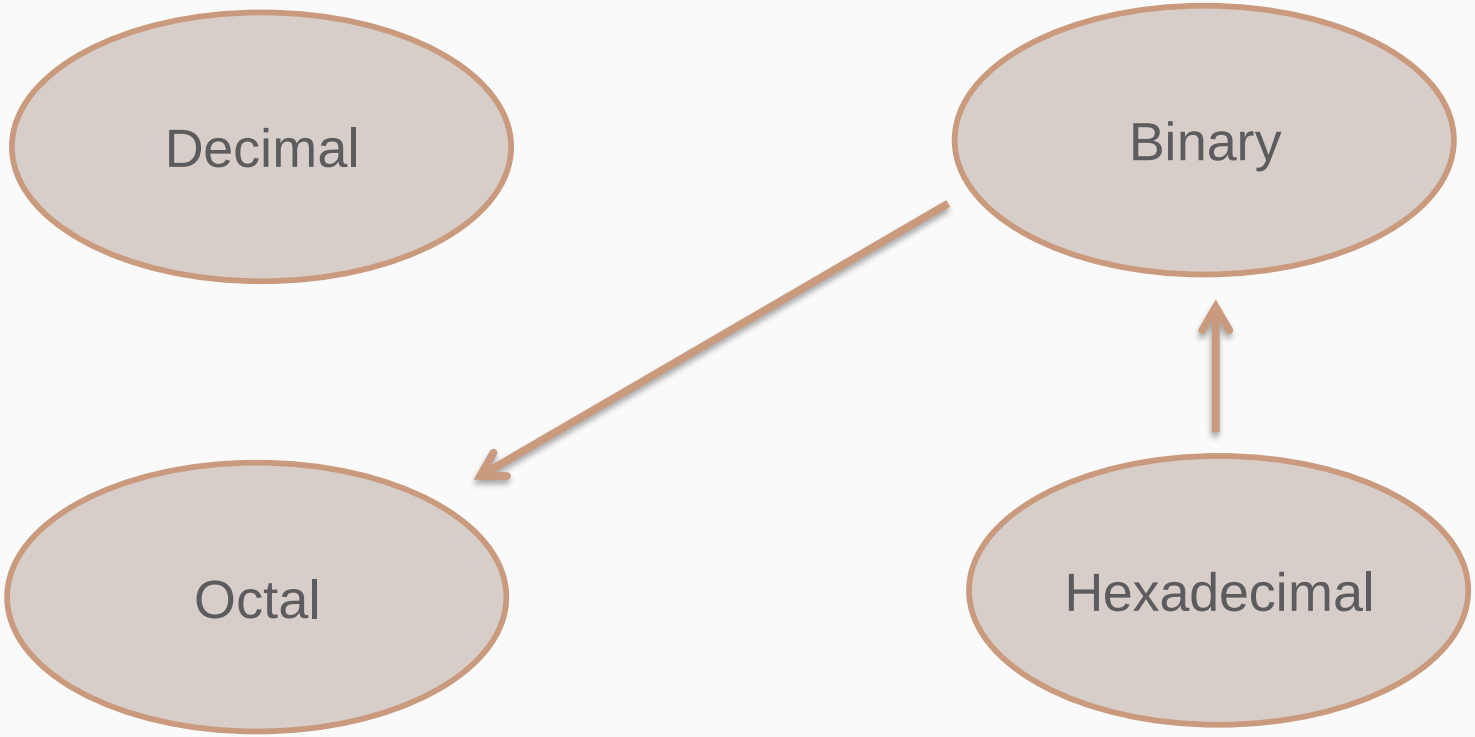
$1076_8 = ?_{16}$



$1076_8 = 23E_{16}$

- Try to solve :
  - $(561)_8$
  - $(3773)_8$
  - $(7765)_8$

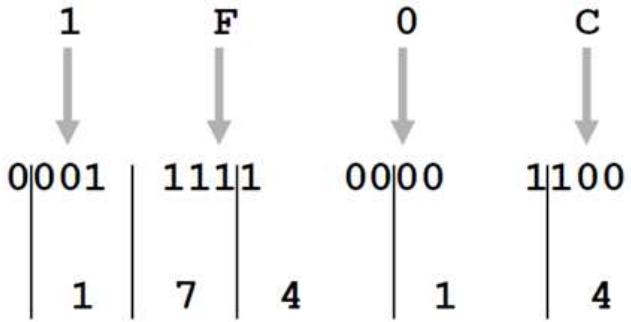
Hexadecimal to Octal



Technique

- Use binary as an intermediary

$1F0C_{16} = ?_8$



$1F0C_{16} = 17414_8$

- Try to solve :
  - $(FCAB)_{16}$
  - $(8FE2)_{16}$
  - $(1100)_{16}$

Thanks!